AIORI HACKATHON 3

THEME: Federated Learning

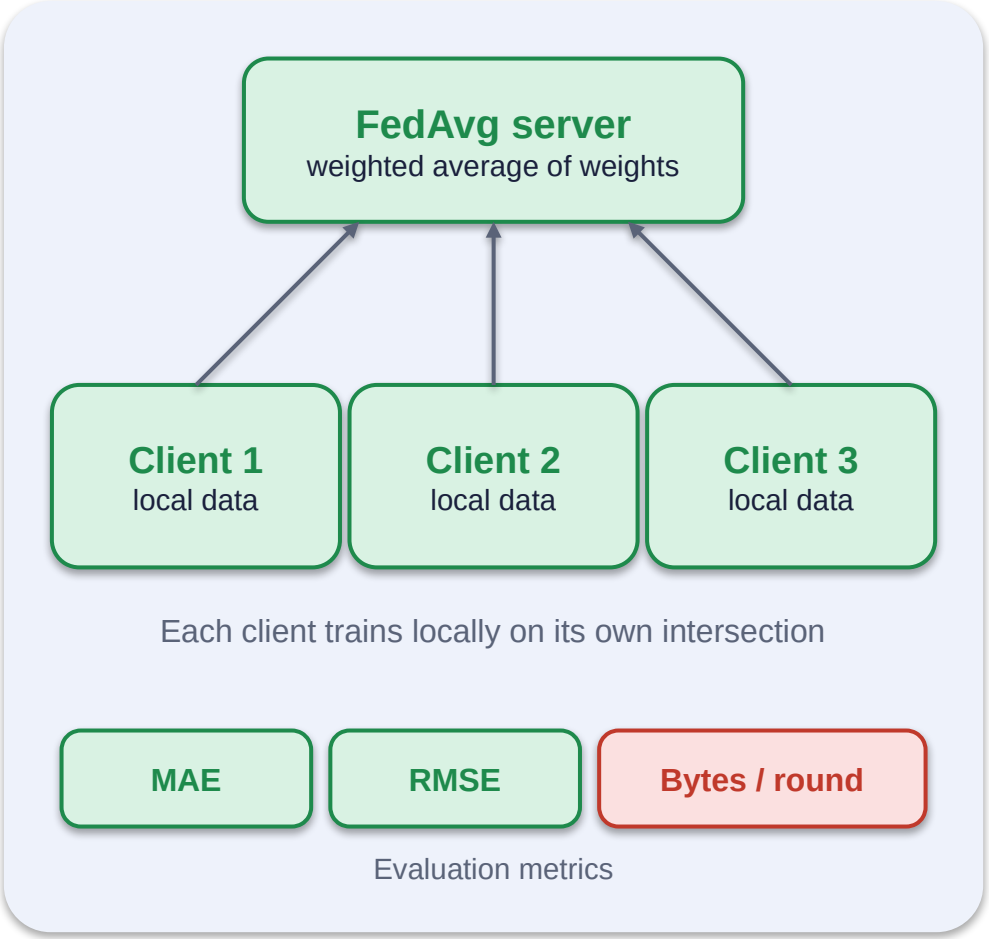
FEDERATED TRAFFIC-FLOW PREDICTION WITH COMMUNICATION-COST EVALUATION

Predicting traffic without pooling raw data.

Team ID: A3-PS003-TC299

Team Name: Error 404

Team Members: Prateek Pandey (Lead), Abhishek, Nitish Sheoran



Problem Description

Predict traffic flow across intersections without pooling raw traffic data.

The need. Traffic-flow forecasting benefits from data of many intersections, but pooling raw records on one server is bandwidth-heavy and exposes sensitive data.

The approach. Federated learning (FedAvg) keeps data on each client, yet real intersections are non-IID: different peak hours, volumes and daily patterns.

Open questions. How close is federated to centralized accuracy, how much does non-IID data hurt, and what does it cost in communication?

PS-003 scope. A public dataset split across 3-5 virtual intersections, chronological train/test split, MAE and RMSE, communication cost, IID vs non-IID, three seeds. Formal privacy guarantees are not required.

Our work. A FedAvg simulator, a centralized baseline, and a research dashboard.

Background

What is needed to understand the problem and the proposed evaluation.

Federated learning: clients train a shared model on local data; only model weights are sent to a server.

FedAvg: each round the server averages client weights, weighted by sample count [1].

Traffic forecasting task: predict the next 5-minute flow from the previous 12 intervals (1 hour) plus time of day.

Non-IID clients: when local distributions differ, the averaged model can lose accuracy [2].

Communication cost: every round each client downloads and uploads the weights, so bytes per round = 2 x clients x model size. Our model has 1,569 float32 parameters (6,276 B), i.e. about 36.8 KB per round for 3 clients.

Metrics: MAE and RMSE (vehicles per interval), bytes per round, rounds to converge.

Literature & RFC Review

Most relevant studies only. No RFC governs this ML simulation study.

Ref	Study	Key idea	Relevance to this work
[1]	McMahan et al., 2017	Federated Averaging (FedAvg)	The aggregation algorithm we implement
[2]	Zhao et al., 2018	Accuracy drops with non-IID client data	Motivates our IID vs non-IID experiment
[3]	Li et al., 2020 (FedProx)	Proximal term for heterogeneous clients	Planned extension for non-IID
[4]	Liu et al., 2020	Federated GRU for traffic-flow prediction	Closest prior work in the traffic domain
[5]	Li et al., 2018 (DCRNN)	Introduced the METR-LA traffic benchmark	Candidate public dataset

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Proposed Solution

A FedAvg simulator with a centralized baseline and a research dashboard.

1 Data

Public 5-minute traffic-flow dataset split into 3-5 virtual intersections; chronological split, first 70% train and last 30% test.

2 Model

Compact MLP 14→32→32→1 (1,569 parameters). Input: last 12 intervals + time-of-day; output: next-interval flow.

3 Federated loop

Server sends global weights; each client trains 2 local epochs (Adam); clients return weights only; server takes the sample-weighted average.

4 Experiments

IID (pooled, shuffled windows) vs non-IID (one intersection per client); 3 seeds; centralized baseline with the same epoch budget.

5 Evaluation

MAE and RMSE on the global test set, bytes per round and in total, rounds to converge (RMSE within 5% of best).

6 Delivery

Python engine (fedtraffic.py), live web dashboard and a Streamlit dashboard.

Optimization Proposed

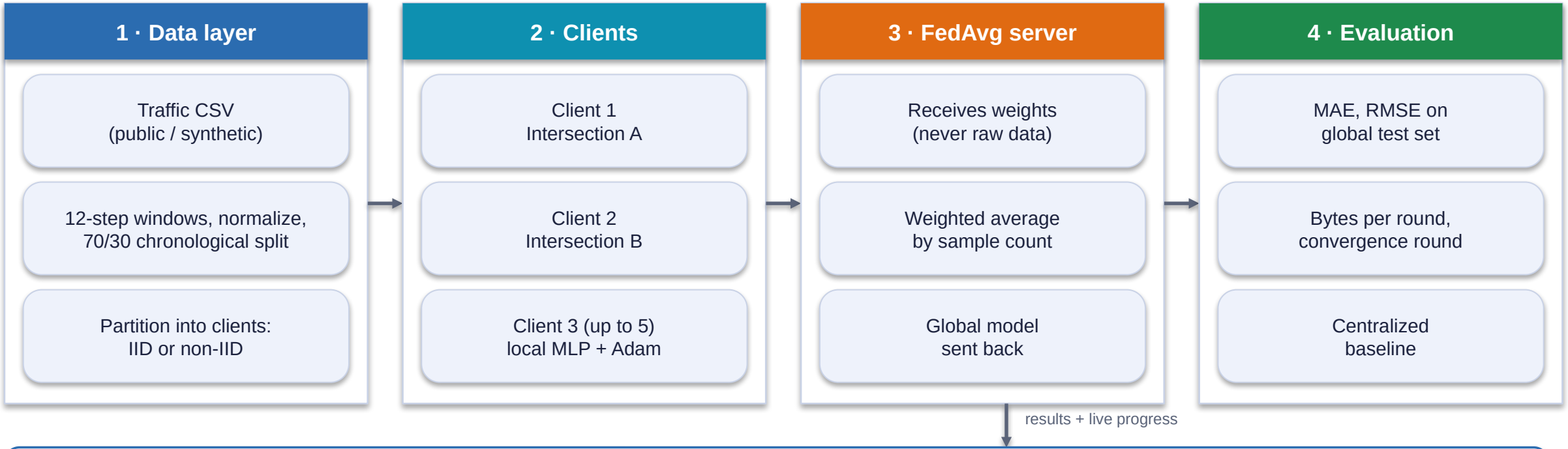
Naive approach vs design choices used in this work.

Area	Before (naive approach)	After (this work)	Benefit
Update size	Large model, many parameters	Compact MLP, 1,569 params (6,276 B)	36.8 KB per round for 3 clients
Aggregation	Plain average of clients	Weighted by sample count	Fair when client sizes differ
Split	Random shuffle of time series	Chronological 70/30 split	No future-data leakage
Baseline	Centralized, arbitrary epochs	Same epoch budget as federated	Fair accuracy comparison
Features	Past values only	+ time-of-day (sin, cos)	Separates morning / evening patterns
Reliability	Single run	3 seeds, mean ± std	Results not one-off luck

Preliminary results (synthetic data, 3 clients, 3 seeds). RMSE: IID federated 4.48 vs centralized 4.54; non-IID federated 5.08 vs centralized 4.35 (about 17% higher). Communication: 36.8 KB per round, converged near round 15-16. Next: FedProx for non-IID, update compression, run on a public dataset.

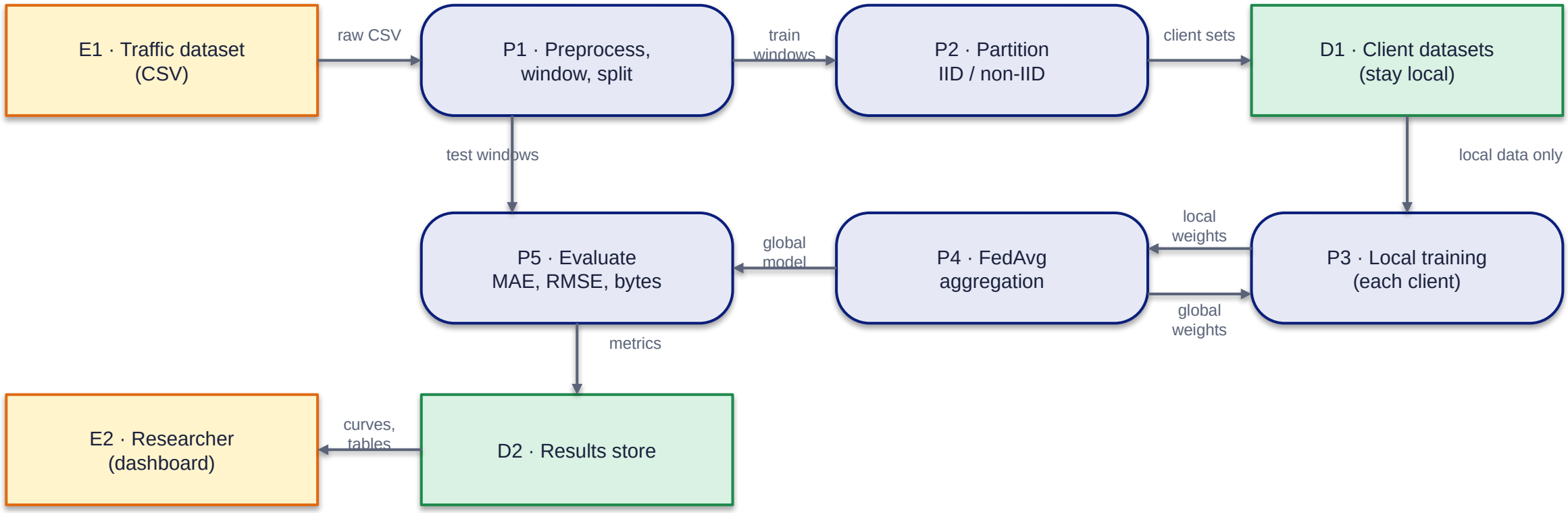
Solution Architecture & Design

Four-layer system: data → clients → FedAvg server → evaluation and dashboard.



Data Flow Diagram

Flow of data between processes, data stores and external entities.



Legend: rectangle (yellow) = external entity · rounded box = process · green box = data store

Deployment Timeline

Milestones and methodology.

Phase	Milestone	Deliverable	Status
1	Dataset module and preprocessing	Windows, chronological split, IID / non-IID partition	Done
2	Forecasting model and centralized baseline	MLP, MAE / RMSE evaluation	Done
3	FedAvg simulator	Weighted aggregation, bytes-per-round counting	Done
4	IID vs non-IID, 3-seed study	Mean ± std table, convergence curves	Done (synthetic data)
5	Dashboards	Web dashboard + Streamlit app, research page	Done
6	Public-dataset run and final report	Results on PeMS / METR-LA, final write-up	Pending

Methodology: iterative build → seeded experiments → comparison with the centralized baseline. Deployment is local (python server.py), with no external services required.

Bibliography

References used in this proposal.

[1] H. B. McMahan et al. “Communication-Efficient Learning of Deep Networks from Decentralized Data.” In: AISTATS (2017).

[2] Y. Zhao et al. “Federated Learning with Non-IID Data.” In: arXiv:1806.00582 (2018).

[3] T. Li et al. “Federated Optimization in Heterogeneous Networks.” In: MLSys (2020).

[4] Y. Liu et al. “Privacy-Preserving Traffic Flow Prediction: A Federated Learning Approach.” In: IEEE Internet of Things Journal (2020).

[5] Y. Li et al. “Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting.” In: ICLR (2018).

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THANK YOU

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Questions & Discussion